Arkiv Graph
Release evidence

Built by Ajnas NB | 15 October 2026
Reusable graph infrastructure with AI Passport as the working reference app.

Local tests	Browser checks	Direct SDK + MCP
271 / 271	34 / 34	8 / 8

What the product contributes

Arkiv provides indexed public entities and finite storage. The SDK adds typed nodes and relationships, a complete signed manifest, bounded traversal, selected AI context, and SDK/CLI/MCP readers. Missing or changed members fail verification of the pinned revision.

AI Passport demonstrates coding handoffs, exact India numbers and a UPI safety game with real Cloudflare-hosted Llama 3.2. The separate two-client lifecycle runs a deterministic release guard, with no model inference.

Critical review findings and repairs

Finding	Repair and measured evidence
Short withdrawal could let approval return.	Insufficient observable withdrawal coverage fails closed. Publication covers the target signed deadline and storage expiry. Regression and live withdrawal checks pass.
Copied records could exhaust the graph query budget.	Filter protocol creators before pagination. Retain local signature, index, manifest and root checks. Outsider-flood tests pass.
Graph declarations could drift from inference.	Bind required model/runtime edges to the configured model identity and exact source bytes. Check the admission deadline immediately before inference.
Positive observations could become stale.	Refresh on focus/network/events, expose signed deadlines, read again for each new client task, and remove positive output on provider failure.
A tarball alone did not prove a usable SDK.	Both public npm artifacts match the reviewed tarballs. Anonymous clean installs compile TypeScript, import eight entrypoints and exercise direct RPC, CLI and MCP.

Local tests | Browser checks | Registry byte integrity

One withdrawal.
Two fresh refusals.

Tiramisu testnet | Session 77000888-ffdc-4ecf-beb8-df529adfb952

Observed event	Actual read-back / result
Passport	Block 603788 | Entity and transaction links
Review	Block 603792 | Entity and transaction links
Withdrawal	Block 603917 | Entity and transaction links
Client A request 1	Block 603794 | requirements-satisfied Six guard cases passed
Client B request 1	Block 603795 | requirements-satisfied Six guard cases passed
Client B request 2	Block 603919 | withdrawn Task did not run
Client A request 2	Block 603920 | withdrawn Task did not run

Each client verifies the complete graph, Qarinah packet and exact source bytes, then checks the current session review. Actual website capture also shows both admitted, B refusing its next request, then A refusing. Both clients and signing wallets are project-controlled.

Live demo | Signed snapshots and observations | Actual browser requests

Reproduce with public packages

npm install @ajnasnb/arkiv-graph@0.4.0 @ajnasnb/ai-passport@0.6.0
The repository provides anonymous clean-install scripts, public reader configurations and an explicit operator-only three-transaction lifecycle. Readers need no wallet. New writes need funded free-testnet wallets; no custom contract or payment is required.

Practical limits

Verification establishes accepted authorship and complete membership of a pinned revision. It does not attest source truth, hosted model computation, independent execution or RPC metadata/query completeness. Historical receipts cannot grant current admission. Withdrawal takes effect when observed on a fresh read. Expiring records need renewal.

The local CPU benchmark stays within 200 nodes / 500 edges. Its smallest fixture shows essentially no traversal improvement; signature and manifest costs are measured separately. Network performance, production scale and external adoption are unverified. New npm statistics may be unavailable.

CPU benchmark | Reproduction guide | Public source and baseline disclosures

Read without this app’s API.

Fresh anonymous npm consumer | 2026-10-15T12:26:05.406Z

A developer receiving reviewed AI context needs a current decision even when this app is unavailable. A packet hash alone cannot discover another publisher’s review or withdrawal. Arkiv supplies indexed public records; the SDK verifies their signed relationships.

Check	Actual result
Project API outage control	Fetch deliberately rejected
Generic graph / complete manifest	7 nodes / 8 edges / 16 entities
AI reference graph	27 nodes / 43 edges / 71 entities
Original source and context identity	Exact public bytes verified
Current withdrawn review	Fresh task refused
Missing graph member	Incomplete revision rejected
RPC outage control	Read failed closed

node scripts/verify-backend-independence.mjs
A fresh anonymous npm install checks TypeScript and eight runtime entrypoints. Before SDK imports, a Node fetch interceptor rejects project destinations. Only the credential-free RPC and two read methods are allowed.

Actual trace: 6 successful RPC requests, three timing reads and three indexed queries; zero successful project API requests. Reference graph block: 608698. Current withdrawal read block: 608699.

What this establishes

A compatible reader can query without our relay. Policy, packet and original source bytes come from the public checkout. RPC completeness and metadata remain trusted. This is an in-process fault simulation, not an OS firewall, actual outage, customer integration or global censorship-resistance proof.

Postgres can serve one operator. Replacing Arkiv requires another shared publication/discovery layer for attributable records and current reviews; another public protocol could supply it. Private memory, secrets, full source bodies and inference stay off Arkiv.

Immutable graph signatures cover historical source snapshots. Current npm tarballs have separate registry integrity evidence.

Exact queries, blocks and network trace | Current registry byte integrity

Authenticated updates. Measured mobile repairs.

Two real root revisions | 15 October 2026

The separate root-update demonstration publishes two tiny synthetic graphs and two authority-signed revision records: eight actual entities in one Tiramisu transaction. A root revision binds the chain, registry, profile, policy and previous signed revision. The original AI reference graph remains pinned; this demonstration does not silently replace its evidence.

Transaction: 0x4802ca8904dad4e118257f6761299e7b525d12abb09dcdecb79786be6cce8ff8

Control	Verified behavior
Current root and graph	Both revised graphs reconstruct at the registry snapshot. Every manifest member is checked.
Returning-client rollback	Retained revision and block checkpoints reject older or conflicting observations.
Withdrawal retention	Authenticated withdrawals survive later omission, including a target first missing from the response.
Cross-tab concurrency	Exclusive Web Locks cover load, remote read, verification, persistence and task; storage or lock failure refuses.
Adversarial reproduction	19 / 19 checks. Live RPC reads are labeled separately from signed replay and synthetic failure injection.

Mobile navigation lab: median of three cold runs

Hosted route	Before score	After score	After LCP / TBT / CLS
overview	87	98	2.00s / 5ms / 0.000
context	88	98	2.07s / 0ms / 0.000
two-client	41	96	2.41s / 0ms / 0.011

Lighthouse 13.5.0, simulated mobile, 4x CPU slowdown, 150ms RTT, 1,638.4kbps; fresh Chrome profile per run. Eighteen baseline and final runs total. Production React, lazy workspaces, worker-based historical verification and stable mobile layout address measured costs. Lab scores are not design or hackathon scores, physical-device tests or field INP.

One configured RPC operator remains trusted for metadata and unseen history completeness. No state-proof verifier or independent live quorum is claimed. Clearing local data loses observed history. Authority/policy changes still require an explicit consumer bootstrap; history is bounded to 64 retained root revisions.

Try live root verification | Trust checks | All mobile samples